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```
#Lowercasing - always the first step
text = ' Hello! This is a SAMPLE Email to check Manoj And Ram. Are looking at screen'
text_lower = text.lower()
print(text_lower)
```

```
hello! this is a sample email to check manoj and ram. are looking at screen
```

```
import re
#Lowercasing - always the first step
text = ' Hello! This is a SAMPLE Email to check Harshi And Nandu. Are looking at screen'
text_lower = text.lower() # Perform lowercasing as per comment
text_clean = re.sub(r'^[a-z]', '', text_lower) # Fix regex and apply to lowercased text
print(text_clean) # Print the cleaned result
```

```
hellothisisasampleemailtocheckharshiandnanduarelookingatscreen
```

```
import nltk
from nltk.corpus import stopwords
nltk.download('stopwords', quiet=True)
stop_words = set(stopwords.words('english'))
text = "I'm going to
tokens = []
for word in text.split():
    if word not in stop_words:
        tokens.append(word)
print(tokens)
```

```
["I'm", 'going', 'college']
```

```
import nltk
nltk.download('punkt_tab', quiet=True)
sentence = "I don't love travelling, do you?"
basic = sentence.split()
print('Basic split:', basic)
```

```
Basic split: ['I', "don't", 'love', 'travelling,', 'do', 'you?']
```

```
from nltk.tokenize import word_tokenize
smart = word_tokenize(sentence.lower())
print('NLTK tokenize:', smart)
```

```
NLTK tokenize: ['i', 'do', "n't", 'love', 'travelling', ',', 'do', 'you', '?']
```

```
import nltk
from nltk.stem import PorterStemmer, WordNetLemmatizer
nltk.download('wordnet', quiet=True) # Corrected package name
stemmer = PorterStemmer()
lemmatizer = WordNetLemmatizer()
Words = ['running', 'studies', 'better', 'caring', 'happily']
for word in Words: # Corrected variable name to 'Words'
    stem = stemmer.stem(word) # Passed individual 'word'
    lemma = lemmatizer.lemmatize(word) # Passed individual 'word'
    print(f'{word:12} | {stem:10} | {lemma}') # Passed individual 'word'
```

running	run	running
studies	studi	study
better	better	better
caring	care	caring
happily	happili	happily

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```

import re
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
nltk.download(['stopwords', 'wordnet', 'punkt'], quiet=True)
stop_words=set(stopwords.words('english'))
lemmatizer=WordNetLemmatizer()
def preprocess(text):
    text=text.lower()
    text=re.sub(r'[^a-z]', '', text)
    tokens=text.split()
    filtered_tokens=[]
    for w in tokens:
        if w not in stop_words:
            filtered_tokens.append(w)
    tokens=filtered_tokens
    lemmatized_tokens=[]
    for w in filtered_tokens:
        lemmatized_tokens.append(lemmatizer.lemmatize(w))
    tokens=lemmatized_tokens
    return ''.join(tokens)
print(preprocess(' I am going out for lunch today'))

```

iamgoingoutforlunchtoday

```

from sklearn.feature_extraction.text import CountVectorizer
emails = [
    'cricket match cricket match cricket'
    'free win money free win money'
    'cricket match free win money cricket'
]
cv = CountVectorizer()
X = cv.fit_transform(emails).toarray()
print('Vocabulary:', cv.get_feature_names_out())
print('Bow matrix')
print(X)

```

Vocabulary: ['cricket' 'cricketfree' 'free' 'match' 'money' 'moneycricket' 'win']
 Bow matrix
 [[3 1 2 3 2 1 3]]

```

from sklearn.feature_extraction.text import TfidfVectorizer
emails = [
    'cricket match cricket match cricket',
    'free win money free win money',
    'cricket match free win money cricket'
]
tfidf = TfidfVectorizer(sublinear_tf=True, max_df=0.95, min_df=1)
X_tfidf = tfidf.fit_transform(emails).toarray()
print('Feature names:', tfidf.get_feature_names_out())
print('TFIDF matrix(rounded):')
import numpy as np
print(np.round(X_tfidf, 3))

```

Feature names: ['cricket' 'free' 'match' 'money' 'win']
 TFIDF matrix(rounded):
 [[0.778 0. 0.628 0. 0.]
 [0. 0.577 0. 0.577 0.577]
 [0.646 0.382 0.382 0.382 0.382]]

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